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What AI can't replace: Rethinking human skills and intelligence

In the rush to embrace AI, it's vital to value our human qualities – and this involves broadening our idea of what education is.

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As artificial intelligence (AI) evolves at breakneck speed, many people worry that machines might soon replace us. But a more important question may be: What exactly are human skills and human intelligence in a world where machines can do so much?

AI is not just automating tasks. It's redefining what we value as "smart" or "skilled".

This shift forces us to reflect not just on employment, but also on education, identity and purpose. It challenges not just how we work, but also how we learn, lead and live alongside machines.

We already see this shift in our daily lives. AI writes software code, generates news summaries and composes music. Tasks that once required formal training or years of creative practice are now done in seconds.

These changes challenge what we consider intelligence, creativity and even human value.

One of the most important lessons of the AI era that has emerged is this: intelligence and wisdom are not the same.

AI can sort data, summarise reports and mimic expression – yet it never pauses, reflects, or weighs long-term consequences.

These are not technical gaps. They are philosophical ones.

Intelligence is knowing how to get somewhere. Wisdom is asking whether it's worth going there. Intelligence solves problems. Wisdom decides which problems matter.

This is why education must shift – from producing skilled coders to nurturing wise citizens.

FROM FIRE-MAKING TO CODING

Skills evolve. Imagine a caveman who could make fire by rubbing sticks together – this was once a vital survival skill. It meant warmth, protection, cooked food, and community. Today, it's a novelty taught at outdoor camps.

Closer to home, many Singaporeans may remember learning woodwork, metalwork or home economics in secondary school. These were essential life skills then.

Today, they've been replaced by design thinking, programming and robotics. As society changes, we redefine what matters.

Even programming – once a mark of technical mastery – is now being reshaped. AI tools can now write functioning code from natural language prompts.

What once took years to master can now be approximated in minutes. This doesn't make programming obsolete, but it forces us to rethink what counts as programming skill: syntax memorisation, or the ability to frame problems, model logic and verify outputs?

NOT JUST REPLACEMENT, BUT RECLASSIFICATION

AI doesn't just replace tasks. It changes how we think about them – what counts as skilled, what counts as intelligent.

Essay writing, translation and medical image reading, once seen as expert work, are now done by machines. What once symbolised intelligence is now outsourced.

In their book The Second Machine Age, authors Erik Brynjolfsson and Andrew McAfee describe how digital technologies are creeping into tasks that once required human thought.

This, they argue, transforms not just jobs, but what society values as intelligence.

So, what remains uniquely human? Not what's hardest to code, but what's hardest to commodify. These are the skills rooted in lived experience, social intuition and moral depth: Emotional, social, moral (integrity, for example), creative, cognitive flexibility (such as perspective-taking) and relational presence, such as accompanying others through uncertainty.

Attendees interacting with a robot at the recent SuperAI conference in Singapore. One of the most important lessons of the AI era that has emerged is this: intelligence and wisdom are not the same, says the writer. AI can sort data, summarise reports and mimic expression – yet it never pauses, reflects, or weighs long-term consequences. PHOTO: BLOOMBERG

These aren't checklist skills. They grow through relationships, responsibility and reflection – not rote learning. In a society like Singapore – diverse in language, culture and belief – these skills build social trust.

Yet in our rush to embrace AI, we often overlook these deeply human capacities. We start to measure ourselves by machine standards: efficiency, output, prediction.

As technologist Jaron Lanier cautions in You Are Not A Gadget: "People degrade themselves in order to make machines seem smart all the time."

He warns that in our desire to celebrate AI, we risk simplifying ourselves – lowering expectations of conversation, judgment, even imagination – to match what machines can mimic.

And once we adjust ourselves to that standard, we begin to believe that machines are smarter, more consistent, even superior. In doing so, we forget the qualities that machines still lack – context, empathy, meaning and presence.

AI can mimic warmth, mirror emotions, speak softly, even offer comfort. But it doesn't grasp what it means to suffer. It doesn't feel fear, regret or grief.

These are not data states. They are human experiences, rooted in vulnerability.

AI is not merely changing what we do, but how we define being human in a world of capable machines. The challenge is then about staying anchored in the values and relationships that make us distinctly human, so that human intelligence is not diminished, but deepened.

HUMAN + AI: AUGMENTED SKILLS, NOT REPLACED ONES

AI need not be a rival. It can be a partner.

Human skills will increasingly include how we prompt, steer and verify AI; how we ensure its outputs serve human goals, reflect ethical judgment and align with context.

At the Singapore University of Technology and Design, where I am based, our Future of Innovation initiative explores precisely this: How AI can be a partner across application contexts, from healthcare to sustainability to education, while emphasising human centricity in each case.

The lessons we draw from these experiments are then brought into the classroom, shaping how students think, not just about using technology, but how and why they use it.

A teacher might use AI to personalise learning. A doctor might use it to catch anomalies. But judgment, empathy and accountability remain human.

Intelligence isn't being replaced. It's being expanded.

The question is whether we expand it with wisdom – by using AI not just to do more, but also to discern what is worth doing.

This means broadening our idea of education to include what is interpretive, ethical, relational and experiential.

AI can't weigh moral complexity. It doesn't know what's right under uncertainty. It doesn't build courage or nurture compassion. These are learnt through culture, mentorship and experience.

True learning isn't just about doing or adapting. It's about discerning and questioning. That kind of education may be the most future-proof of all.

Psychologist Howard Gardner's definition of intelligence is apt: "An intelligence is the ability to solve problems, or to create products, that are valued within one or more cultural settings."

By that standard, intelligence isn't a test score. It's contextual. As context shifts, so must what we value.

THE HUMAN FUTURE

As AI takes on more tasks, we may realise that the most irreplaceable skills are not technical, but moral, social and imaginative. Not what helps us compete with machines, but what helps us remain meaningfully human.

Singapore's National AI Strategy speaks of empowering lives through AI. That vision will be strongest when it is paired with a deeper commitment to what machines must never do for us: feel, choose, or care on our behalf.

These are not just emotional or social capacities. They are the foundations of moral life.

Among them, choosing is especially vital. It is the exercise of agency – the ability to weigh consequences, act with intention and take responsibility.

And agency is the root of human dignity: Our worth lies not just in what we do, but also in the fact that we can choose how and why we do it. To cede that role, even gradually, is to erode the very ground of freedom, ethics and personhood.

Singapore is known for planning ahead. The goal shouldn't be to future-proof jobs. It should be to prepare people for a world where jobs will change, disappear and reappear.

What must remain is our ability to adapt with compassion, creativity and courage.

Perhaps human intelligence was never about doing more – but about understanding what matters most.

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